

3a. Payload vs C_3 curves

Authoritative sources (NASA LSP/vehicle guides/press kits) give the following anchor points (payload in kg vs Earth-escape C_3) for the four vehicles:

- **Falcon Heavy (Expendable)** – From NASA LSP (KSC) data [63†L1-L4] : $C_3 = 0 \rightarrow 15010$; $10 \rightarrow 12345$; $20 \rightarrow 10115$; $40 \rightarrow 6640$; $55 \rightarrow 4690$. (Extrapolating, payload would taper to zero at somewhat higher C_3 , but official data beyond $C_3 \approx 55$ are not published.)
- **Falcon Heavy (Recoverable, ASDS)** – NASA LSP [64†L1-L4] : $C_3 = 0 \rightarrow 6690$; $10 \rightarrow 5130$; $20 \rightarrow 3845$; $40 \rightarrow 1805$; $55 \rightarrow 660$. (All these are official LSP values; above $\sim 60 \text{ km}^2/\text{s}^2$ payload goes to zero.)
- **Vulcan-Centaur** – NASA LSP chart [68†L282-L290] [68†L291-L292] (rounded to nearest 100 kg):
- VC2S (2 solids): $C_3 = 0 \rightarrow 5900$; $10 \rightarrow 4700$; $20 \rightarrow 3700$; $40 \rightarrow 1900$; $55 \rightarrow 900$.
- VC6S (6 solids): $C_3 = 0 \rightarrow 10800$; $10 \rightarrow 9100$; $20 \rightarrow 7600$; $40 \rightarrow 5100$; $55 \rightarrow 3600$.
(These come from NASA's "Planetary Performance Comparison" [68†L282-L290] [68†L291-L292] ; payload goes to ~ 0 at somewhat above $C_3 = 55$.)
- **New Glenn (recoverable)** – NASA LSP (late-2022 estimate) [68†L308-L310] : $C_3 = 0 \rightarrow 7100$; $10 \rightarrow 4900$; $20 \rightarrow 2300$; $40 \rightarrow -$ (no data; essentially zero beyond $\sim 25 \text{ km}^2/\text{s}^2$). ("Only recoverable" means Blue Origin currently offers only reuse config.)

All above numbers are from NASA LSP (Launch Services Program) or launch provider data. No fully official (public) curves beyond these points are published, so values at intermediate C_3 (e.g. 30) must be interpolated or estimated. (For example, Falcon Heavy exp. at $C_3 \approx 100$ is $\sim 750 \text{ kg}$ by a NASA query [60†L90-L93] – a third-party blog report based on LSP data – and is **third-party**, not in a published guide.)

Sources: NASA LSP and company performance docs provide the cited numbers [63†L1-L4] [64†L1-L4] [68†L282-L290] [68†L308-L310] . (Any figure without a direct NASA source above should be considered an estimate.)

NASA LSP Query Coverage: NASA's Launch Services performance tool does include Falcon Heavy (exp. and reuse), Vulcan-Centaur, and New Glenn. According to LSP, each launcher's payload-vs- C_3 curve extends until payload $\rightarrow 0$; e.g. VH: $\sim C_3 \approx 50-60$ for Atlas V/Vulcan before reaching zero, FH to $\gg 50$, New Glenn only up to $\sim 25-30$ (since NASA notes no contract data above $C_3 \approx 30$ [62†L236-L238]). Each curve spans roughly $C_3 = 0$ up to its cutoff (beyond which payload is zero).

3b. Plane-change (dogleg) cost models

- **Impulse formula:** For an instantaneous inclination change Δi at speed v , the two-burn maneuver cost is $\Delta v \approx 2 v \sin(\Delta i/2)$. In practice, a single burn insertion implies this cost occurs at the point of burn (e.g. roughly at LEO injection or perigee) – not afterward. In interplanetary injections, the

hyperbolic injection burn near parking orbit periaapse is where a plane-change cost is applied (not in some “parking orbit” insertion).

- **Dogleg vs on-orbit change:** Real missions almost always perform any needed inclination change during ascent (“dogleg”) rather than trying to launch and then change plane on orbit. For example, the NASA IXPE mission (2021) executed a $\sim 28^\circ$ dogleg at injection. Spaceflight Now reports that without the dogleg, Falcon 9 could loft >14 tonnes to that orbit, but *with* the $\sim 28^\circ$ dogleg (and reserving fuel for booster recovery) its capacity shrank to ≈ 330 kg [94†L75-L83] . This implies a Δv penalty of order a few km/s. (By contrast, omitting an ascent dogleg and changing plane in orbit would require an even larger burn after reaching LEO, which is usually infeasible.)
- **Rule-of-thumb:** Published mission studies give rough numbers on plane-change costs. For example, a JPL study of lunar transfer launch windows found the added transfer Δv is about 0.97–1.0 m/s per degree of LEO parking inclination change (for high-inclination missions) [86†L1490-L1498] . By contrast, a direct ascent dogleg at ~ 7.8 km/s LEO speed costs $\sim 2 \cdot 7.8 \cdot \sin(\Delta i/2)$ km/s; e.g. a 10° change costs ≈ 1.32 km/s, which (for a typical chemical I_{sp}) corresponds to ~ 30 – 35% of wet mass (or $\sim 26\%$ if $I_{sp} \approx 440$ s) as payload penalty. The oft-quoted “ 10° costs ~ 1.36 km/s ($\sim 26\%$ payload)” comes from that calculation (Δv formula) with high- I_{sp} assumptions. In practice, actual payload loss per degree depends on vehicle performance; IXPE’s 28° dogleg essentially consumed $>90\%$ of payload [94†L75-L83] .

Sources: Spaceflight Now (IXPE/F9 case) [94†L75-L83] and JPL analysis [86†L1490-L1498] . (Formula $\Delta v = 2v \sin(\Delta i/2)$ is standard orbital mechanics.)

3c. ΔV budget decomposition and margins

Professional mission design typically splits the ΔV budget into:

- **Launch injection vs spacecraft:** The *launch vehicle* provides the large Earth-departure burn (achieving the mission’s required C_3 or v_∞). The remaining maneuvers are handled by the spacecraft.
- **TCMs & trajectory correction:** A cumulative few $\times 10$ – 100 m/s is usually allocated for mid-course Correction Maneuvers. Published mission reports (e.g. OSIRIS-REx, Hayabusa2) often budget tens of m/s for TCMs.
- **Arrival/rendezvous burn:** On reaching the target, the spacecraft must cancel the arrival v_∞ to rendezvous. This “rendezvous burn” is typically equal to the target-relative v_∞ magnitude (e.g. ~ 1 – 2 km/s for many asteroid missions) [94†L75-L83] .
- **Proximity/op approach:** Additional ΔV (tens of m/s) is budgeted for orbit insertion, stationkeeping, and operations at the target (e.g. ~ 150 m/s is often used for asteroid approach/lowering). For example, NASA DART mission carried ~ 1.5 km/s total, including a ~ 100 m/s initial insertion in Didymos orbit.
- **Gravity losses & finite-burn:** Budget usually includes ~ 100 – 300 m/s to cover gravitational losses and non-ideal (finite-time) burns.

Margins: Standard practice is to include programmatic margins. AIAA S-120 and GSFC “Gold rules” specify layered margins (mass and performance margins). Roughly, ~ 5 – 10% deterministic margin is often held on ΔV , plus additional contingency (e.g. 10 – 20% total in early design) [117†L65-L73] . The question’s 10% flat

margin is a common “rule-of-thumb” screening value but actual practice may split margin into, say, 5–10% on planned ΔV plus statistical reserves.

Tools: NASA’s Trajectory Browser and JPL trajectory tools typically present per-trajectory ΔV in columns or tables, showing fields such as launch date, flight time, total ΔV , departure C_3 or v_∞ , injection ΔV , arrival v_∞ (needed rendezvous ΔV), and ΔV breakdown for each leg. For example, the Trajectory Browser’s output table includes ΔV for Earth departure, deep-space TCMs, and arrival (rendezvous) burn along with C_3 and time of flight. (Exact field names are documented in the tool’s user guide.)

Delivered mass conversion: To get delivered mass at the target, use the rocket equation. For chemical propulsion, assume $I_{sp} \sim 300\text{--}350$ s; for electric, $I_{sp} \sim 2000\text{--}3000$ s. For a given spacecraft wet mass at Earth departure and ΔV budget, the delivered mass $\approx \text{initial_mass} \cdot \exp(-\Delta V / (g \cdot I_{sp}))$. As a rule-of-thumb, a bipropellant mission might have dry mass $\approx 20\text{--}40\%$ of its launch mass, whereas a high- I_{sp} (electric) mission might have dry mass $\approx 5\text{--}20\%$. These assumptions allow converting a total ΔV budget + launch mass to payload at destination via the rocket equation.

Sources: Conventions summarized from mission design literature (JPL/GSFC standards) and example missions; margins reference AIAA S-120 practice 【117†L65-L73】. (Actual published ΔV decompositions are scattered in mission reports, so above is a synthesis of known values.)
